Newton’s inverse square law for gravity (Chapters 2 and 23) forces the orbits of the planets around the Sun to be ellipses. Each planet feels small gravitational tugs from the other planets, and these tugs cause its ellipse to gradually change orientation, that is, to gradually *precess*.

In 1859, the astronomer Urbain Le Verrier at the Observatoire de Paris (France) announced he had discovered an anomaly in the orbit of the planet Mercury. When he computed the total precession of Mercury’s orbit caused by all the other planets, he got the wrong answer. The measured precession is larger than the planets could produce by about 0.1 arc second each time Mercury traverses its orbit (Figure 24.1).

Now 0.1 arc second is a tiny angle, just one ten-millionth of a circle. But Newton’s inverse square law insists there can be no anomaly whatsoever.

Le Verrier convinced himself that this anomaly is produced by the gravitational tug of an undiscovered planet closer to the Sun than Mercury; “Vulcan” he called it.

Astronomers searched in vain for Vulcan. They could not find it, nor could they find any other explanation for the anomaly. By 1890 the conclusion seemed clear: Newton’s inverse square law must be very slightly wrong.

Wrong in what way? A revolutionary way, it turned out. The way discovered by Einstein twenty-five years later. The warping of time and space endow the Sun with a gravitational force that obeys Newton’s inverse square law, but only nearly. Not precisely.

Upon realizing that his new relativistic laws explain the observed anomaly, Einstein was so excited that he suffered heart palpitations and felt like something snapped inside himself. “For a few days I was beside myself with joyous excitement.”